

Powered Autonomous Delivery Aircraft (PADA)

Final Project Technical Report

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1 Conceptual Design

1.1 Mission Requirements and Design Constraints

The primary design goal for the Powered Autonomous Delivery Aircraft (PADA) is to carry out an autonomous delivery mission while strictly adhering to physical, aerodynamic, and structural constraints [?, ?]. The governing design constraints for the system are defined as follows:

- **Maximum Wingspan ($b_{\max}$):** Restricted to **25 inches** (with a maximum landing gear width limit of 20 inches) [?].
- **Maximum Gross Weight ($W_{\max}$):** Capped at **19.5 oz** (≈ 552.8 g/5.421 N) including all avionics and structural components [?, ?].
- **Wing Planform Area (S):** Standardized at 0.08 m^2 , resulting in a baseline wing loading (W/S) of 67.76 N/m^2 [?, ?].
- **Payload Specification:** An external empty Pepsi can secured and released via a dedicated payload release mechanism [?].
- **Battery & Energy Limit:** Total electrical energy aboard the aircraft must remain strictly $\leq 100 \text{ Wh}$ [?, ?].
- **Stability & Control:** The aircraft must exhibit static and dynamic stability across longitudinal, lateral, and directional axes [?, ?].

1.2 Initial Aircraft Configurations Considered

During the initial design phase, multiple conventional and non-conventional UAV layouts were evaluated against payload delivery feasibility, structural efficiency, and ease of component integration:

- **Conventional Tractor Monoplane (Selected Baseline):** Preferred for high propeller efficiency, straightforward longitudinal trimming, and optimal mass distribution for a nose-mounted motor combined with an external ventral payload bay.
- **Twin-Boom Pusher Configuration:** Evaluated to clear the forward field of view for optical sensors; however, it was discarded due to structural mass penalties from twin booms exceeding the $W_{\max}$ budget of 552.8 g [?].
- **High-Wing Placement:** Chosen over low-wing concepts to ensure high lateral stability (pendulum effect) and maximum ground clearance for the external payload release mechanism during landing operations.

1.3 Selected Configuration, Layout, and Airfoil Selection

1.3.1 Airfoil Selection Trade Study

A formal decision matrix was conducted to select the optimal wing profile among three candidate airfoils evaluated at $Re = 200,000$ using XFLR5 [?, ?]: **E423**, **GOE165**, and **S1223** [?]. The selection criteria and weightings were defined as: $C_{L,\max}$ ($W_1 = 0.35$), $(L/D)_{\max}$ ($W_2 = 0.25$), $C_{D,\min}$ ($W_3 = 0.20$), and C_{m0} ($W_4 = 0.20$) [?, ?].

Table 1: Airfoil Selection Decision Matrix ($Re = 200,000$)

Parameter	Weight (W_i)	Raw Ratings (s_{ij})			Weighted Scores ($W_i \cdot s_{ij}$)		
		E423	GOE165	S1223	E423	GOE165	S1223
$C_{L,\max}$	0.35	0	-1	+1	0.00	-0.35	+0.35
$C_{D,\min}$	0.20	-1	+1	0	-0.20	+0.20	0.00
$(L/D)_{\max}$	0.25	+1	-1	0	+0.25	-0.25	0.00
C_{m0}	0.20	0	+1	-1	0.00	+0.20	-0.20
Total Score (S_j)	1.00	–	–	–	+0.05	-0.20	+0.15

The **Selig S1223** achieved the highest total score (+0.15) and was selected as the wing profile to maximize takeoff performance and high-lift capabilities under tight wing loading constraints [?, ?].

1.3.2 Wing Taper Ratio Selection

Evaluating taper ratios (λ) at $V_{\text{cruise}} = 20$ m/s in XFLR5 demonstrated that $\lambda = 0.6$ provided optimal aerodynamic performance, yielding a maximum lift-to-drag ratio ($(L/D)_{\max} = 14.25$) alongside a strong negative pitching moment slope ($\partial C_m / \partial \alpha$) [?].

1.4 Major Design Trade-offs and Justification

- Airfoil High-Lift vs. Pitching Moment (S1223 vs. Trim Drag):** Although the S1223 induces a large negative pitching moment ($C_{m0} = -0.2716$), its superior maximum lift coefficient ($C_{L,\max} = 2.2314$) is essential for short takeoff ground rolls [?, ?]. The high pitching moment is balanced by sizing a horizontal tail volume ratio of $V_{HT} = 0.7$ with an optimal tail arm of $L_{\text{opt}} = 0.311$ m [?].
- Avionics Mass vs. Structural Margin Optimization:** Initial baseline electrical configurations (3S 1300 mAh battery) totaled 359 g, leaving only 193.8 g for the airframe [?]. Downsizing to an **850 mAh LiPo** reduced mass by 40 g, providing necessary structural weight allowance while operating safely within the < 100 Wh limit [?, ?].

2 Preliminary Design

2.1 Propulsion Constraint Analysis and Sizing

Based on $W_{\max} = 19.5$ oz ≈ 5.421 N and wing area $S = 0.08$ m² ($b = 0.635$ m), operational wing loading is calculated as $W/S = 67.76$ N/m² [?, ?]. Evaluating flight

constraint equations yields minimum thrust-to-weight targets: $(T/W)_{\text{Climb}} \approx 0.22$ and $(T/W)_{\text{TO}} \approx 0.38$ for an 8 m ground roll [?]. Applying a 20% safety margin for efficiency losses sets the required static performance at $(T/W)_{\text{req}} \geq 0.50$ ($T_{\text{req}} \geq 276.4$ g) [?].

Tail and control surfaces were sized using volume coefficient methods ($V_{\text{HT}}, V_{\text{VT}}$) and historical ratios [?]. Horizontal Tail (HT) trade study evaluated taper options ($0.4T$ – $1.0T$), selecting $0.6T$ for optimal balance between aerodynamic efficiency and static stability slope ($C_{m\alpha}$) [?]. Primary control surfaces (ailerons, elevator, rudder) were dimensioned and integrated into the CAD assembly [?].

2.2 Mass Budget and CG Distribution

Component mass breakdown yields a baseline total system mass of 514 g (well within the 552.8 g ceiling), establishing an initial Center of Gravity at $X_{\text{CG}} \approx 0.068$ m [?]:

Table 2: System Mass Budget and Longitudinal CG Breakdown

Component	Mass (g)	X-Loc (m)	Component	Mass (g)	X-Loc (m)
Main Wing	125	0.073	Motor	28	-0.050
Fuselage & H/W	134	0.080	Avionics & Elect.	124	0.050–0.090
Horiz. Tail (HT)	18	0.454	Battery (850 mAh)	75	0.010
Vert. Tail (VT)	10	0.448	Total System	514 g	~ 0.068 m

2.3 Aerodynamic Modeling and Stability (AVL / XFLR5)

The geometric model was evaluated in AVL at $V = 20.0$ m/s ($Re \approx 200,000$) to verify trim performance and dynamic stability derivatives [?]. At the trimmed operating point ($\alpha = -8.303^\circ$), AVL yielded $C_L = 0.25723$, $C_D = 0.01696$ ($C_{Di} = 0.016958$), and span efficiency $e = 0.2679$ [?]. Stability derivative analysis confirmed static longitudinal stability ($C_{m\alpha} < 0$) and stable dynamic modes, validating the configuration prior to detailed assembly [?, ?].

3 Detailed Design

3.1 Aircraft Geometry and Structural Assembly

The aircraft geometry was designed within the strict physical constraints defined for the Powered Autonomous Delivery Aircraft (PADA) project. The wing planform utilizes a low aspect ratio configuration ($AR = 5.04$) with a uniform chord distribution ($c = 0.20$ m).

The structural layout incorporates balsa wood ribs and carbon fiber main spars for high bending stiffness, integrated with an Expanded Polypropylene (EPP) fuselage structure and 3D-printed payload mounting components.

3.2 Final Mass Estimation and CG Location

The overall mass budget was constrained below the maximum threshold of $W_{\text{max}} = 552.8$ g. The component breakdown and Center of Gravity (CG) locations measured

Table 3: PADA Aircraft Geometry Parameters

Parameter	Value	Unit
Wingspan (b)	25.0	in (0.635 m)
Wing Area (S)	0.08	m ²
Root Chord (c_{root})	0.20	m
Tip Chord (c_{tip})	0.20	m
Aspect Ratio (AR)	5.04	-
Airfoil Section	Selig S1223	-
Horizontal Tail Area (S_{ht})	0.016	m ²
Vertical Tail Area (S_{vt})	0.009	m ²

relative to the wing leading edge (LE) are detailed in Table 4.

Table 4: Final Mass Estimation and Center of Gravity Location

Component / Subsystem	Mass (g)	CG Position X (m)
Main Wing Assembly	125.0	0.073
Fuselage and Structural Hardware	134.0	0.080
Horizontal Tail (HT)	18.0	0.454
Vertical Tail (VT)	10.0	0.448
LiPo Battery (Tattu 3S 850mAh)	75.0	0.010
Motor (SunnySky X2204 KV2300)	28.0	-0.050
Avionics & Payload Release Electronics	124.0	0.050 – 0.090
Total Gross Weight	514.0	0.073 (24.2% MAC)

3.3 Final Aerodynamic and Stability Verification

Aerodynamic and static longitudinal stability verification was conducted using AVL and XFLR5 computational toolsets. The selected Selig S1223 high-lift airfoil yields a maximum lift coefficient ($C_{L,max} = 2.2314$) under low Reynolds number conditions ($Re = 200,000$).

Linear vortex-lattice modeling confirms a stable pitching moment slope ($C_{m,\alpha} < 0$) with the neutral point located at 33% MAC. Operating at the design Center of Gravity location (24.2% MAC) ensures a positive static margin of approximately 8.8%, guaranteeing longitudinal static stability across the mission flight envelope.

4 Electrical and Avionics Design

4.1 Components Inventory

4.2 Master Electrical Schematic Diagram

Detail the wiring, signal routing, and power distribution schema.

Table 5: PADA Electrical Components Inventory

Component	Specs / Rating	Price	Weight (g)	Selected Model
Battery				
ESC				
Motor				
Flight Controller				
Receiver				
GPS				
Telemetry				
Servos				

5 Autonomous System and Software

5.1 MAVROS and ArduPilot SITL Integration

Explain software setup, MAVROS nodes, and SITL simulation loop.

5.2 Visual Perception for Autonomous Mission Adaptation

Detail computer vision pipelines, detection models, and precision target tracking.

5.3 Mode-Switching Logic and Path Planning

Explain flight mode transition state machines and path-planning algorithms.

6 Payload Release Mechanism

6.1 Mechanical Design and Triggering Logic

Describe external payload release mechanism, servo actuators, and release logic.

6.2 Material Selection

6.2.1 Introduction

This trade study evaluates five candidate materials for the primary structural components of a radio-controlled (RC) plane [?]. The materials are graded on a scale of +1 (Good), 0 (Acceptable/Neutral), and -1 (Poor) across three criteria: Viability (performance and suitability), Price (cost-effectiveness), and Manufacturability (ease of building and shaping) [?].

6.2.2 Trade Study Matrix

The detailed comparison matrix and evaluation scores for all candidate materials are summarized in Table 6 [?].

Table 6: Grading Matrix for RC Plane Materials

Material	Viability	Price	Manufacturability	Total Score
Balsa Wood	+1	+1	+1	+3
Foam (EPP/Depron)	0	+1	+1	+2
3D-Printed (LW-PLA)	0	0	0	0
Plywood (Lite/Birch)	0	0	-1	-1
Carbon Fiber	+1	-1	-1	-1

6.2.3 Detailed Evaluation

- **Balsa Wood (+3):** The traditional gold standard for model aircraft [?]. It offers an excellent strength-to-weight ratio, is relatively inexpensive, and is exceptionally easy to cut, sand, and glue [?]. It provides the best balance of all criteria [?].
- **Foam (+2):** A highly cost-effective and easy-to-work-with alternative [?]. It is excellent for crash resistance and rapid prototyping, though it often requires structural reinforcement (like carbon rods) to achieve sufficient rigidity for flight [?].
- **3D-Printed (+0):** Offers incredible design freedom for complex parts using lightweight filaments like LW-PLA [?]. However, it is a slow manufacturing process, requires a high initial printer investment, and produces more brittle parts compared to traditional materials [?].
- **Plywood (-1):** While stronger and stiffer than balsa, plywood is significantly heavier [?]. It is best used sparingly for high-stress areas like firewalls rather than a full airframe [?]. It is also harder to cut and shape [?].
- **Carbon Fiber (-1):** The highest-performance material, offering unmatched strength and stiffness [?]. However, it is extremely expensive and difficult to manufacture safely, requiring specialized tools and protective equipment due to hazardous dust [?].

7 Final Assessment and Bonus Work

7.1 Flight Simulation and Performance Validation

Present flight test simulation logs and performance evaluation against requirements.

7.2 Bonus Features Implementation

Document additional innovative features or bonus system implementations.

References

1. Gudmundsson, S. *General Aviation Aircraft Design: Applied Methods and Procedures*.